O-S-34B
December 14, 1970
SUPERSEDING
Fed. Spec. O-S-34A
June 24, 1953

FEDERAL SPECIFICATION

SODIUM NITRATE, TECHNICAL (NITRATE OF SODA)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a commercial (technical) grade of sodium nitrate.

1.2 Classification. Sodium nitrate shall be furnished in one type and grade.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids, or request for proposal form a part of this specification to the extent specified herein.

Federal Specifications:

UU-S-48 - Sacks, Shipping, Paper.
PPP-D-705 - Drum: Metal Shipping, Steel (Over 12 and Under 55 Gallon).

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Test Method Std. No. 536 - Soap and Soap-Products (Including Synthetic Detergents); Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies).

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads 40" x 48" 4-Way (Partial) Pallet Skids, Runners, or Pallet-Type Base.

FSC 6810

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Laws and Regulations:

49CF100 to 199 - Hazardous Materials Board, Department of Transportation.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

Association of Official Analytical Chemists (AOAC) Publications:

Methods of Analysis-Tenth Edition.

(Application for copies should be addressed to P. O. Box 540, Benjamin Franklin Station, Washington, DC 20044.)

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies shall be addressed to the National Motor Freight Traffic Association, Inc., Agent, 1616 P Street N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, IL 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Material. Sodium nitrate shall either be a natural or synthetic product conforming to the requirements of table I.

3.2 Workmanship. The quality of workmanship shall be such as to produce sodium nitrate that is in accordance with the requirements of this specification.

TABLE I. Chemical and physical requirements

Purity of $NaNO_3$, percent by weight (min.)	97
Granule size, percent passing through USS sieve size	
No. 4 min.	100
No. 30 max.	10
Moisture content and other volatile matter	
Percent by weight (max.)	0.5

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection lot. A lot shall consist of sodium nitrate manufactured in one day with the same raw materials and the same manufacturing process.

4.2.1 Unit of product. A unit of product shall be one container of sodium nitrate.

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, Special Inspection Level S-2. Approximate equal portions of sodium nitrate shall be taken from each sample container, composited, and transferred directly to a clean, thoroughly dried bottle labeled to identify the lot and containers from which it was taken. The bottle shall be immediately tightly stoppered. The total composited sample shall be no less than one pound. The composited sample shall be used for tests specified in 4.4. All tests shall be performed in duplicate. If the samples fail to meet the requirements of table I, the lot shall be rejected.

4.3 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing and marking comply with the requirements of section 5. Defects shall be scored as specified in table II. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

TABLE II. Examination of preparation for delivery

Examine	Defect
Containers	Not as specified.
Contents	Not as specified.
Markings	Omitted, incorrect, illegible, improper size location, sequence, or method of application.
Materials	Component missing or damaged.
Workmanship	Bulging or distortion of containers. Cushioning inadequate, improper or missing.

4.4 Tests.

4.4.1 Sodium nitrate purity. The sodium nitrate purity shall be determined by the Devarda method or by the Ferrous-Sulfate-Zinc-Soda method specified in the Official Methods of Analysis issued by the Association of Official Analytical Chemists, (AOAC).

4.4.2 Particle size test. The samples selected at random as specified in 4.2.2 shall be tested in accordance with Fed. Test Method 536/2101 to determine the size of the sodium nitrate particle.

4.4.3 Moisture content. Transfer a 10 gram (g.) sample of sodium nitrate to a previously tared weighing bottle. Heat the bottle and sample in an oven for 2 hours at 302° + 5°F. (150° + 3°C.). After 2 hours, cover bottle, remove from oven, cool in desiccator and weigh. Calculate the percentage of moisture in the sample as follows:

$$\text{Percent moisture} = \frac{A - B}{W} \times 100$$

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Where: A = Wt. (in g.) of bottle plus sample before drying.
B = Wt. (in g.) of bottle plus sample after drying.
W = Wt. (in g.) of sample before drying.

4.5 Inspection of preparation for delivery. The packing and marking of the sodium nitrate shall be inspected to determine conformance with the applicable requirements of section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Sodium nitrate shall be packed in metal drums conforming to PPP-D-705, type III. Gross weight of containers shall not exceed 120 pounds. Containers shall be provided with 0.004 inch polyethylene liners, sealed by heat sealing.

5.1.2 Level B. Sodium nitrate shall be packed in shipping sacks conforming to the applicable level B requirements of UU-S-48.

5.1.3 Level C. Sodium nitrate shall be packed in a manner that will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.2 Palletization. When specified, sodium nitrate packed in accordance with 5.1 shall be palletized in accordance with MIL-STD-147 (see 6.2).

5.3 Marking. In addition to any special marking required by contract or order, shipping containers shall be marked in accordance with Fed. Std. 123 or MIL-STD-129, as applicable (see 6.2).

5.4 Compliance with statutory regulations. Packing and marking shall conform to the Department of Transportation rules and regulations for the Transportation of Hazardous Materials, 49CFR100 to 199.

6. NOTES

6.1 Intended use. Sodium nitrate covered by this specification is intended for use in salt baths, and for the heat treatment of metals. For other uses see 6.3.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Level of packing required (see 5.1).
(c) Marking required (see 5.3).
(d) Palletization, if required (see 5.2).

6.3 Sodium nitrate for pyrotechnics and other ordnance applications is covered by MIL-S-322.

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MILITARY INTEREST:

Custodians:
Army - MU
Navy - YD
Air Force - 68

User activity:

Army - CE

Preparing activity:

Navy - YD

CIVIL AGENCY INTEREST:

GSA-FSS

Orders for this publications are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

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SPECIFICATION VALIDATION RECORD

Form Approved
OMB No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by anyone, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for the submittal of comments on the use of this specification which will insure that quality products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Feel free to use an additional sheet(s) as necessary. Comments on this specification submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

SPECIFICATION: Federal Specification O-S-634
Sodium Nitrate, Technical (Nitrate of Soda).

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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